

Customer Churn Analysis Report

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Customer Churn Analysis

Understanding Why Customers Leave and How to Improve Customer Retention

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Role: Data Analyst

Tools Used:

- Python
- Pandas
- Matplotlib
- Jupyter Notebook
- Power BI

Project Type: Exploratory Data Analysis (EDA)

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Executive Summary

Customer retention is one of the biggest challenges for every telecom company. Losing customers not only reduces revenue but also increases the cost of acquiring new customers. The purpose of this project is to understand why customers leave the company and identify the factors that influence customer churn.

In this project, I analyzed customer data using Python and performed data cleaning, exploratory data analysis, and business analysis. Several customer characteristics such as contract type, payment method, internet service, monthly charges, and customer tenure were studied to understand their relationship with churn.

The analysis shows that customers with month-to-month contracts, higher monthly charges, electronic check payment methods, and shorter tenure are more likely to leave the company. Based on these findings, practical recommendations have been provided to help improve customer retention and reduce churn.

Business Problem

The telecom company is experiencing customer churn, which is affecting revenue and long-term business growth.

The company wants to answer the following question:

Why are customers leaving the company, and what actions can be taken to reduce customer churn?

Understanding the reasons behind customer churn will help the company improve customer satisfaction and increase customer retention.

Project Objectives

The main objectives of this analysis are:

- Understand the overall customer churn rate.
 - Identify customer groups with higher churn.
 - Analyze how different customer characteristics affect churn.
 - Discover the major reasons behind customer churn.
 - Provide business recommendations to improve customer retention.
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Dataset Overview

The dataset contains customer information collected by a telecom company.

Dataset Information

- Total Customers: 7,043
- Total Features: 21
- Target Variable: Churn

The dataset includes information such as:

- Customer demographics
 - Internet services
 - Contract details
 - Payment methods
 - Monthly charges
 - Total charges
 - Customer tenure
 - Customer churn status
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Data Cleaning

Before starting the analysis, the dataset was carefully checked for quality issues.

The following data cleaning steps were performed:

- Checked missing values.
- Checked duplicate records.
- Verified data types.
- Converted data into appropriate formats where required.
- Removed missing records.
- Saved the cleaned dataset for further analysis.

These steps helped ensure that the analysis was performed using reliable and accurate data.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand customer behavior and identify patterns related to churn.

The following business questions were analyzed:

- What is the overall customer churn rate?
- Does contract type affect churn?
- Does internet service affect churn?
- Does payment method affect churn?
- Does gender influence churn?
- Do senior citizens churn more?
- Does having a partner reduce churn?
- Do customers with dependents stay longer?
- Does Tech Support reduce churn?
- Does Online Security reduce churn?
- Does Paperless Billing affect churn?
- Are customers with higher monthly charges more likely to churn?
- Does customer tenure affect churn?

Bar charts, pie charts, and box plots were used to visualize customer behavior and identify meaningful business insights.

Key Findings

The analysis revealed several important findings.

- Around one out of every four customers has left the company.
- Customers with month-to-month contracts have the highest churn.
- Customers using Fiber Optic internet service are more likely to leave.

- Electronic Check users show higher churn than customers using automatic payment methods.
 - Customers paying higher monthly charges are more likely to churn.
 - Customers with shorter tenure leave the company more frequently.
 - Customers with Tech Support are more likely to stay.
 - Customers with Online Security have better customer retention.
 - Senior citizens have a relatively higher churn rate.
 - Customers with partners and dependents generally stay longer with the company.
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Business Recommendations

Based on the analysis, the following recommendations are suggested:

- Encourage customers to switch from month-to-month contracts to yearly contracts by offering discounts or loyalty rewards.
- Improve the quality and customer experience of Fiber Optic services.
- Promote automatic payment methods through cashback or promotional offers.
- Offer bundled plans that include Tech Support and Online Security.
- Provide loyalty rewards for customers with high monthly charges.
- Improve the onboarding experience for new customers during their first year.
- Develop special retention programs for senior citizens and high-risk customer groups.

These actions can help reduce customer churn and improve long-term customer satisfaction.

Business Impact

Implementing these recommendations can provide several business benefits.

- Reduce customer churn.
- Improve customer satisfaction.
- Increase customer lifetime value.
- Reduce customer acquisition costs.
- Improve long-term revenue.
- Build stronger customer relationships.

Even a small reduction in churn can have a significant positive impact on the company's profitability.

Conclusion

This project successfully analyzed customer churn using real business data.

The analysis identified several factors that influence customer churn, including contract type, payment method, monthly charges, customer tenure, internet service, and additional support services.

The insights and recommendations provided in this report can help the company make informed business decisions and improve customer retention.

This project demonstrates the complete workflow of a data analyst, from understanding the business problem to providing actionable business recommendations based on data.